FINITENESS VIA PROPERNESS AND FINITE FIBRES OVER A LOCALLY NOETHERIAN BASE: FOR F : X -> S WITH S LOCALLY NOETHERIAN, F IS FINITE IF AND ONLY IF F IS PROPER WITH FINITE FIBRES (RETAINED WITH THE AUTHOR'S PARENTHETICAL POINTING TO THE MORE GENERAL MORE ON MORPHISMS TWIN).

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6097-6108 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 6110-6149, the complete author proof (38 body lines).

Editorial note. Navigation stubs: 10 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool stacks-coherent backfill wave 33c. precard_gate RESULT=OK (probe L9999).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-characterize-finite' also exists in stacks-spaces-more-morphisms — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 02OE: sources/stacks-coherent/pinned-original-source.tex, source lines 6013–6021. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01XB: sources/stacks-coherent/pinned-original-source.tex, source lines 363–367. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01XG: sources/stacks-coherent/pinned-original-source.tex, source lines 608–611. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02LS: sources/stacks-coherent/pinned-original-source.tex, source lines 6149–6149. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01WJ: sources/stacks-coherent/pinned-original-source.tex, source lines 6149–6149. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01WN: sources/stacks-coherent/pinned-original-source.tex, source lines 6149–6149. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02NU: sources/stacks-coherent/pinned-original-source.tex, source lines 6149–6149. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01WM: sources/stacks-coherent/pinned-original-source.tex, source lines 6149–6149. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02NH: sources/stacks-coherent/pinned-original-source.tex, source lines 6149–6149. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01LC: sources/stacks-coherent/pinned-original-source.tex, source lines 6149–6149. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *(For a more general version see More on Morphisms, Lemma 02LS.) Let $f : X \to S$ be a morphism of schemes. Assume S is locally Noetherian. The following are equivalent*

(1) *f is finite, and*
(2) *f is proper with finite fibres.*

Proof. A finite morphism is proper according to Morphisms, Lemma 01WN. A finite morphism is quasi-finite according to Morphisms, Lemma 02NU. A quasi-finite morphism has finite fibres, see Morphisms, Lemma 02NH. Hence a finite morphism is proper and has finite fibres.

Assume f is proper with finite fibres. We want to show f is finite. In fact it suffices to prove f is affine. Namely, if f is affine, then it follows that f is integral by Morphisms, Lemma 01WM whereupon it follows from Morphisms, Lemma 01WJ that f is finite.

To show that f is affine we may assume that S is affine, and our goal is to show that X is affine too. Since f is proper we see that X is separated and quasi-compact. Hence we may use the criterion of Lemma 01XG to prove that X is affine. To see this let $\mathcal{I} \subset \mathcal{O}_X$ be a finite type ideal sheaf. In particular $\mathcal{I}$ is a coherent sheaf on X. By Lemma 02OE we conclude that $R^1 f_* \mathcal{I}_s = 0$ for all $s \in S$. In other words, $R^1 f_* \mathcal{I} = 0$. Hence we see from the Leray Spectral Sequence for f that $H^1(X, \mathcal{I}) = H^1(S, f_* \mathcal{I})$. Since S is affine, and $f_* \mathcal{I}$ is quasi-coherent (Schemes, Lemma 01LC) we conclude $H^1(S, f_* \mathcal{I}) = 0$ from Lemma 01XB as desired. Hence $H^1(X, \mathcal{I}) = 0$ as desired. □